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Weekly update 5 of 5 · week of Dec 7

Features, LEED v5 and deliverables

THIS UPDATE IN THREE NUMBERS

10

early-design features are as good as all 48

7×

prompt content moved the error 7× more than any feature set

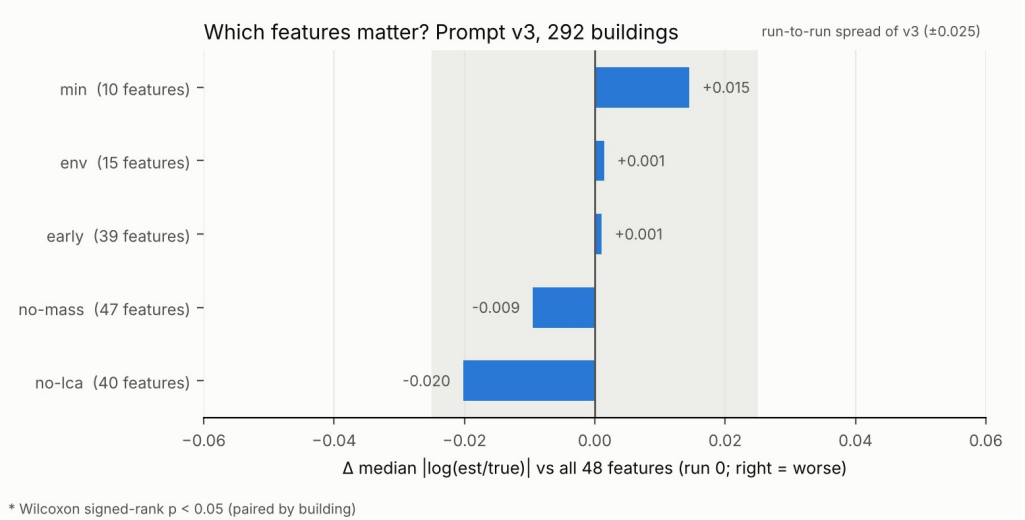
≈ 6%

of whole-building carbon from a 30% enclosure cut

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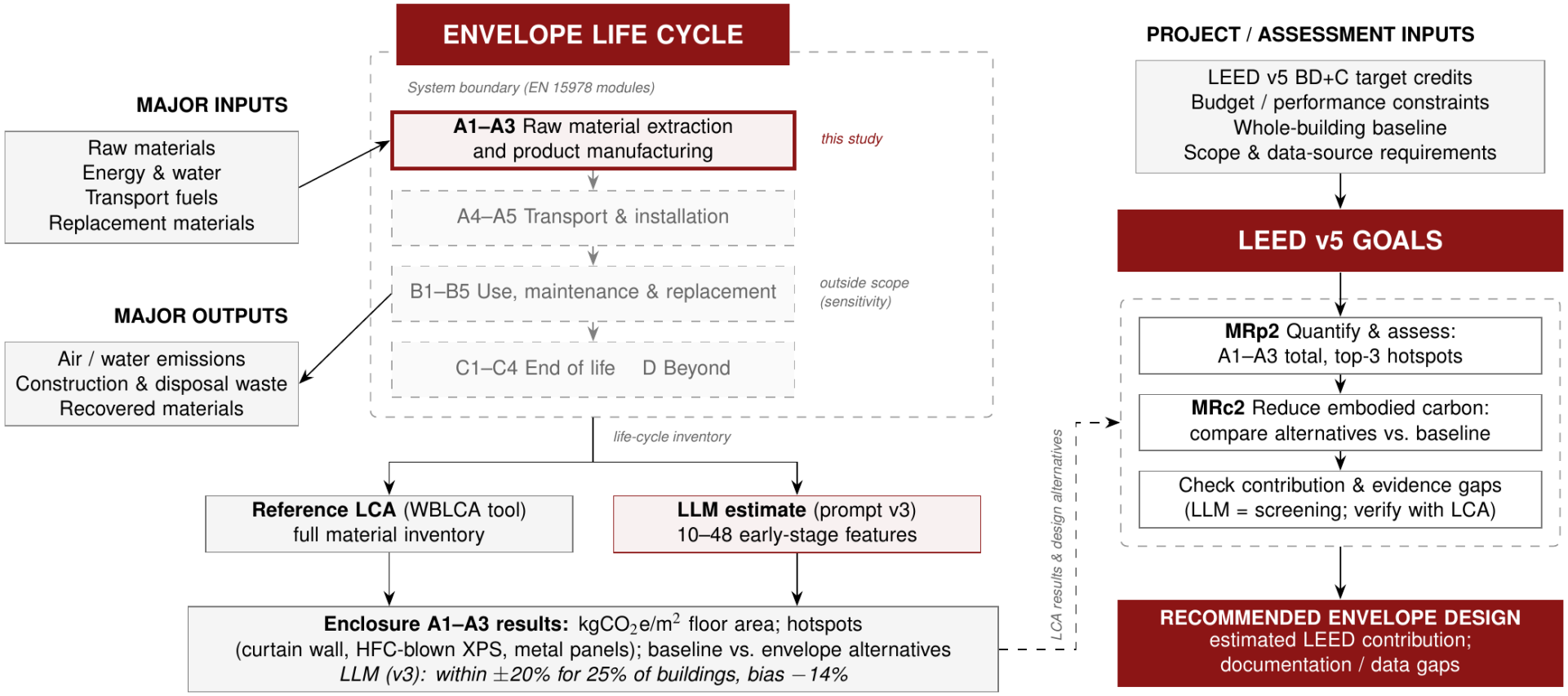
Feature study: 10 inputs are enough (v3, run 0)

Set	n	Coverage	Med. |log|	Δ	p	Bias
all	48	24.3%	0.461	–	–	0.86
early	39	21.6%	0.462	+0.001	0.13	0.88
env	15	19.9%	0.463	+0.001	0.33	0.93
min	10	25.3%	0.476	+0.015	0.81	0.81
no-lca	40	25.3%	0.441	−0.020	0.57	0.90
no-mass	47	22.6%	0.452	−0.009	0.92	0.85



- No set differs significantly from all 48; every difference is within v3's run-to-run spread (±0.025).
- The model does **not depend on mass_total** or other LCA-derived fields.
- One run per set; 7 of 9 planned ablations were stopped to limit cost.

Where the estimate fits in LEED v5



The study boundary is A1–A3. A reference LCA and the LLM estimate both produce enclosure results that feed **MRp2** (quantify, hotspots) and **MRc2** (reduce against a baseline). The LLM screens; an LCA documents.

LEED v5: screening yes, credit documentation no

1 MRp2: hotspots

Bias is small (−14%; near zero without XPS), and the reasoning names the real drivers: curtain wall and framing, HFC-XPS, metal panels, glazing. **Suited to schematic-design screening.**

2 MRc2: reductions

CV 12.5%, median error ≈ 35–40%, coverage 1 in 4: as large as typical documented reductions. **Cannot document a credit-level reduction.** Paired comparison of alternatives is the promising, untested use.

3 Option 1: dilution

The enclosure is a median **19%** of whole-building A–C, so a 30% enclosure cut ≈ **6%** of the building. Envelope choices work best together with structure, or via Option 2 EPDs.

Deliverables and limitations

1 Final prompt (v3)

Full text in the paper's appendix, with the sourced intensity table.

2 Paper, reports, poster

ICML-style paper, English and Chinese reports, PPT poster; all code and results in the GitHub repo.

3 Limitations

Answers in one task file share a context; post-hoc split; v3–v1 gain borderline with 3 runs; feature study one run; per m² floor area; uneven reference quality; one model.

4 Future work

Assembly-type inputs; paired alternative comparison; hotspot-agreement test; an external test set.

Wrap-up

Take-aways

- An LLM's error is **structured**: compression and material blind spots, which error analysis can locate.
- **Sourced numbers at the decision step** work; qualitative knowledge does not.
- For LEED v5: **screening and comparison**, verified by an LCA.

Input we'd like

- Feedback on the paper and poster before final submission.
- Interest in the **assembly-type** follow-up?